

Automobile Sales Array Algorithms

Part (a): Find the number of years with sales > 300

Algorithm CountYearsOver300(AUTO)

```
1. SET NUM = 0
2. FOR K = 1932 TO 1984 DO
3.   IF AUTO[K] > 300 THEN
4.     NUM = NUM + 1
5.   ENDIF
6. ENDFOR
7. PRINT "Number of years with more than 300 cars sold =", NUM
END Algorithm
```

Part (b): Print each year and the number of cars sold

Algorithm PrintYearSales(AUTO)

```
1. FOR K = 1932 TO 1984 DO
2.   PRINT "Year =", K, " Cars Sold =", AUTO[K]
3. ENDFOR
END Algorithm
```

Part (c): Combined Algorithm (Searching + Reporting)

Algorithm CombinedSalesReport(AUTO)

```
1. SET NUM = 0
2. FOR K = 1932 TO 1984 DO
3.   PRINT "Year =", K, " Cars Sold =", AUTO[K]
4.   IF AUTO[K] > 300 THEN
5.     NUM = NUM + 1
6.   ENDIF
7. ENDFOR
8. PRINT "Total number of years with more than 300 cars sold =", NUM
END Algorithm
```